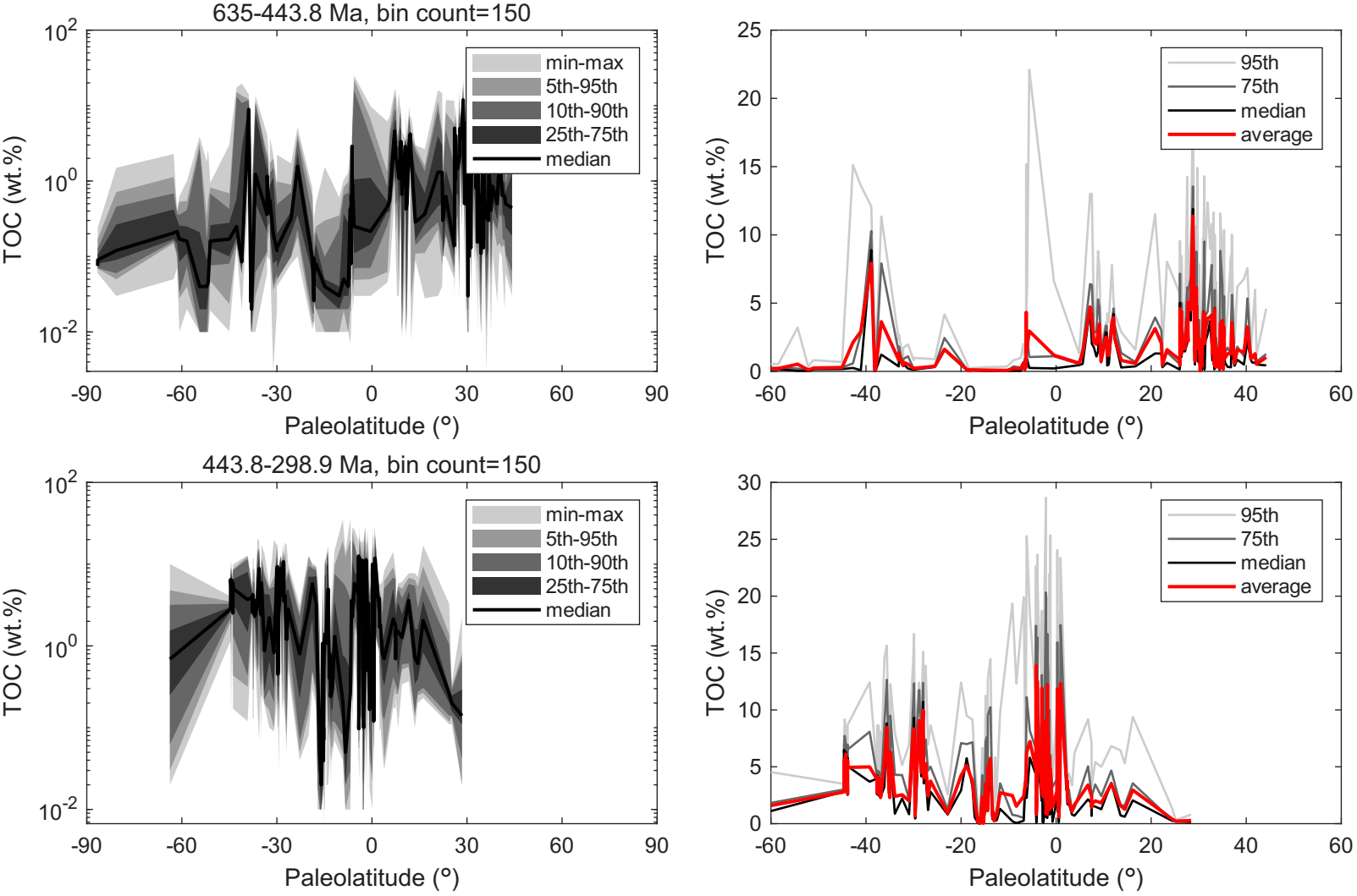
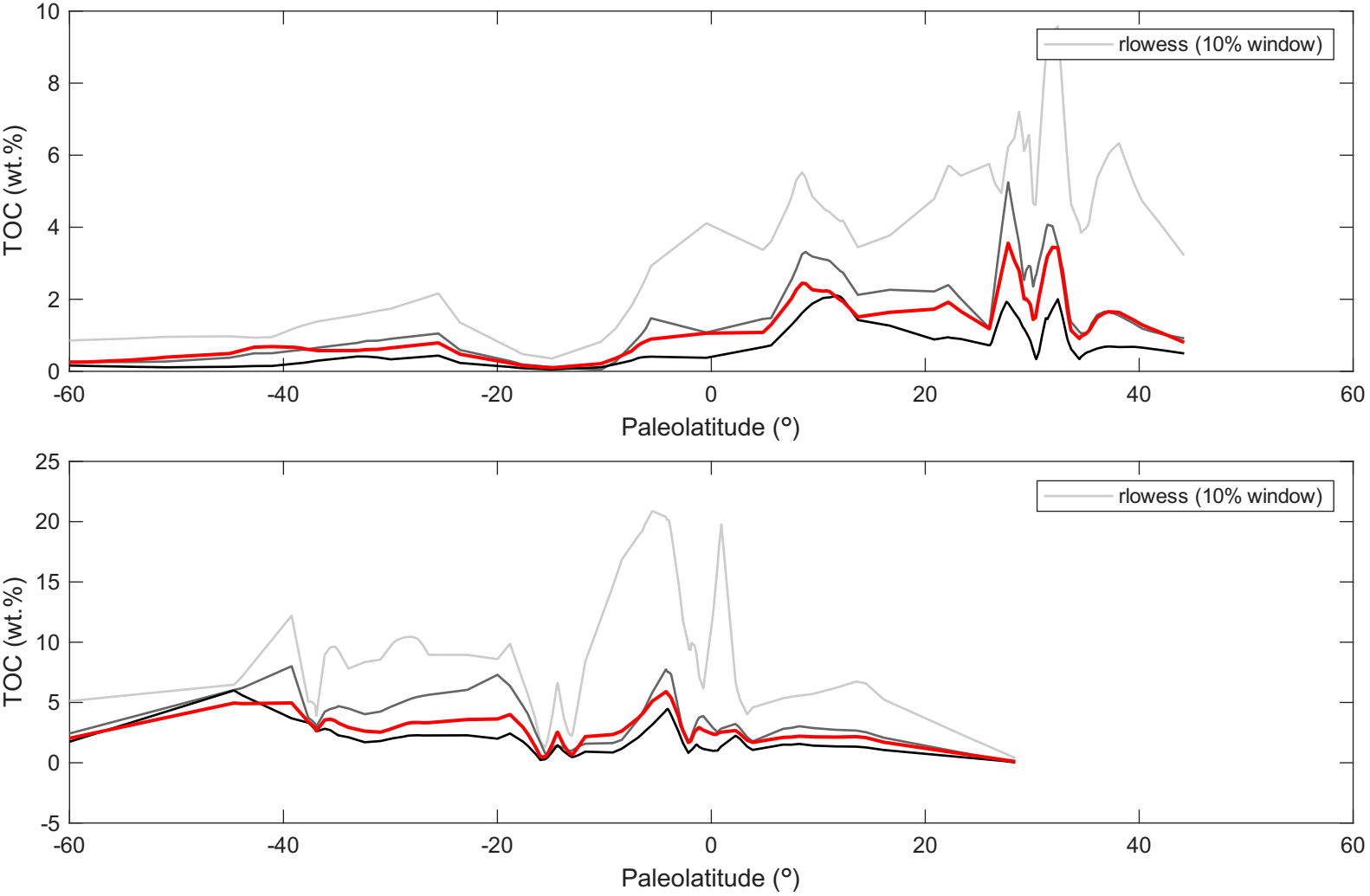
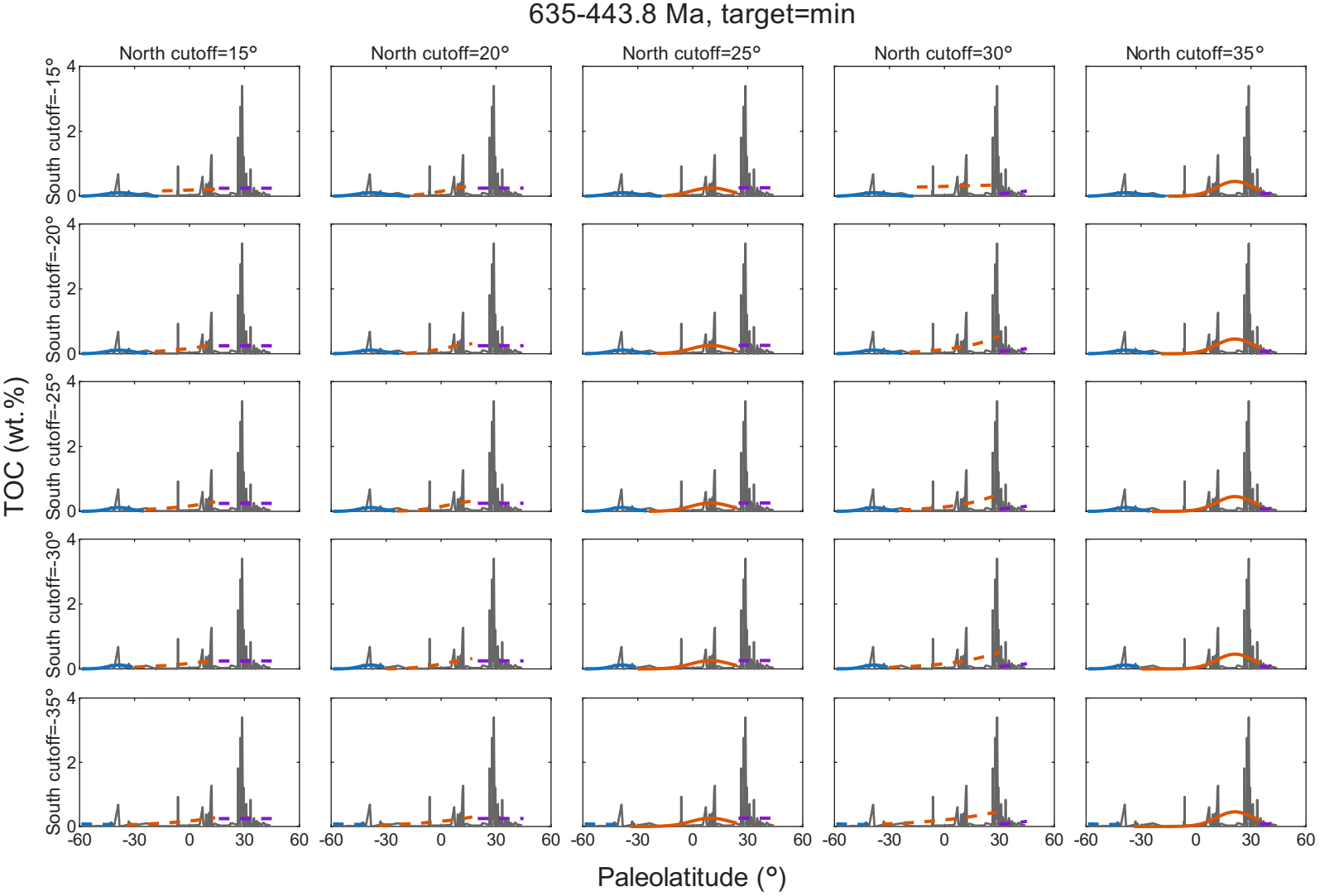
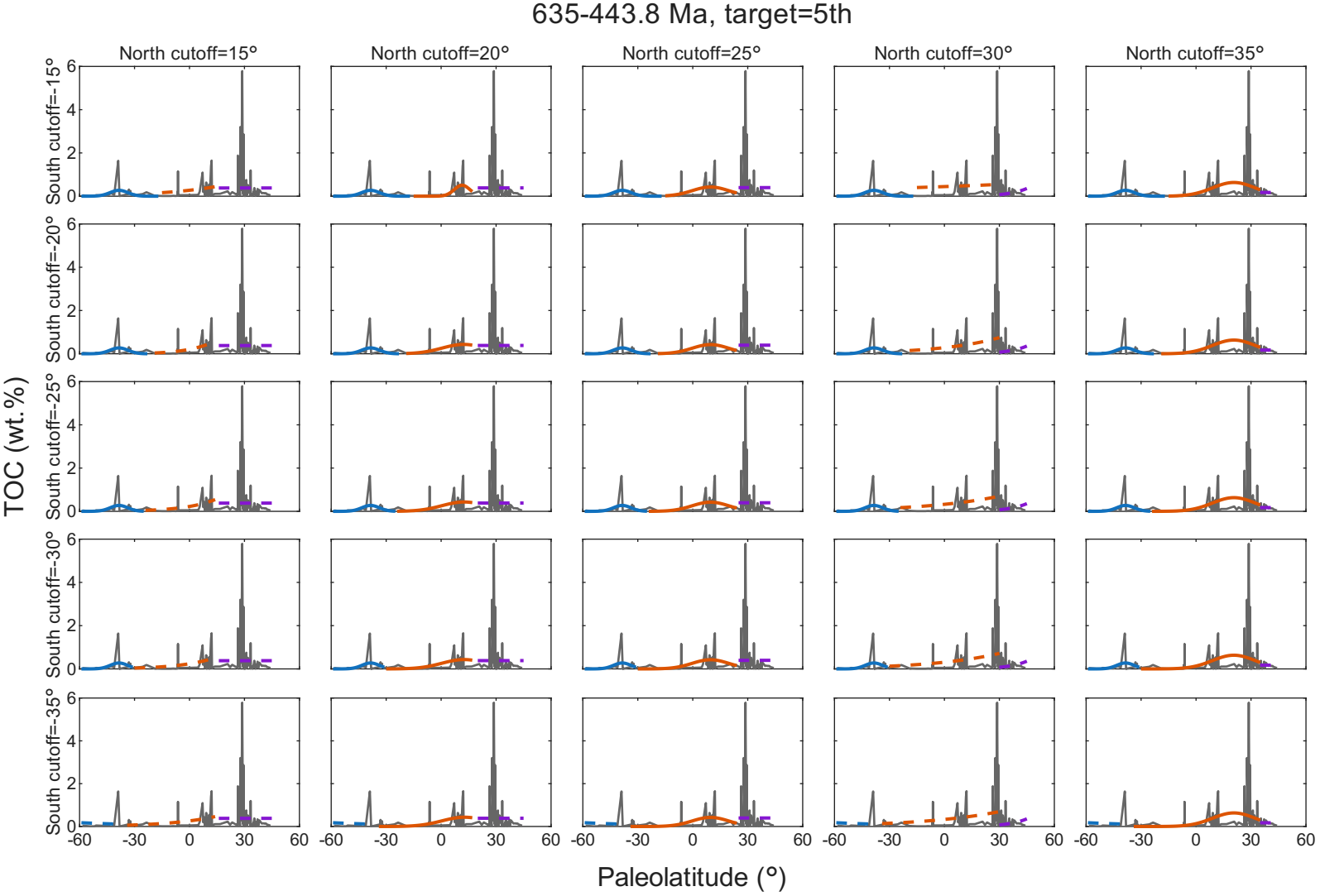
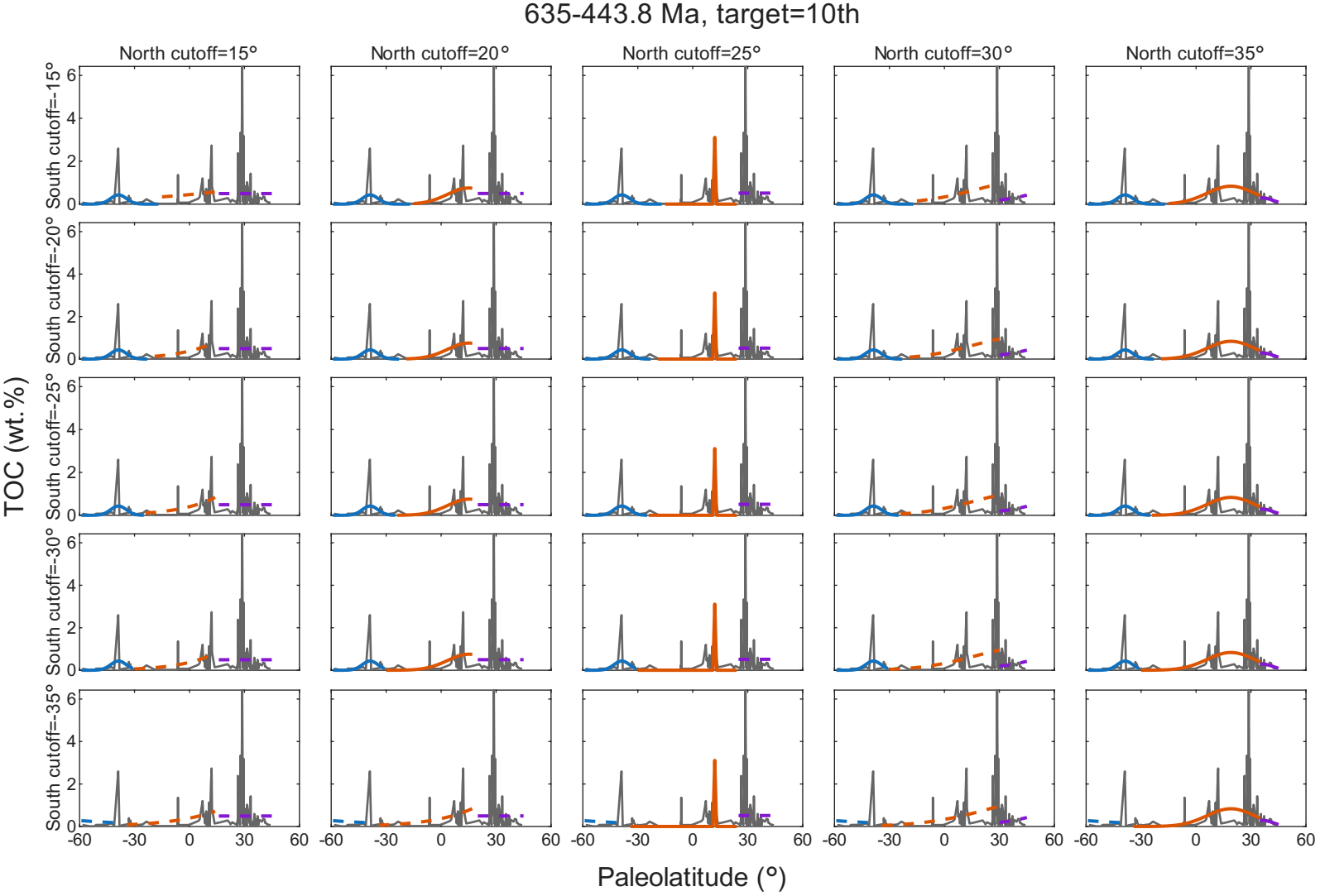
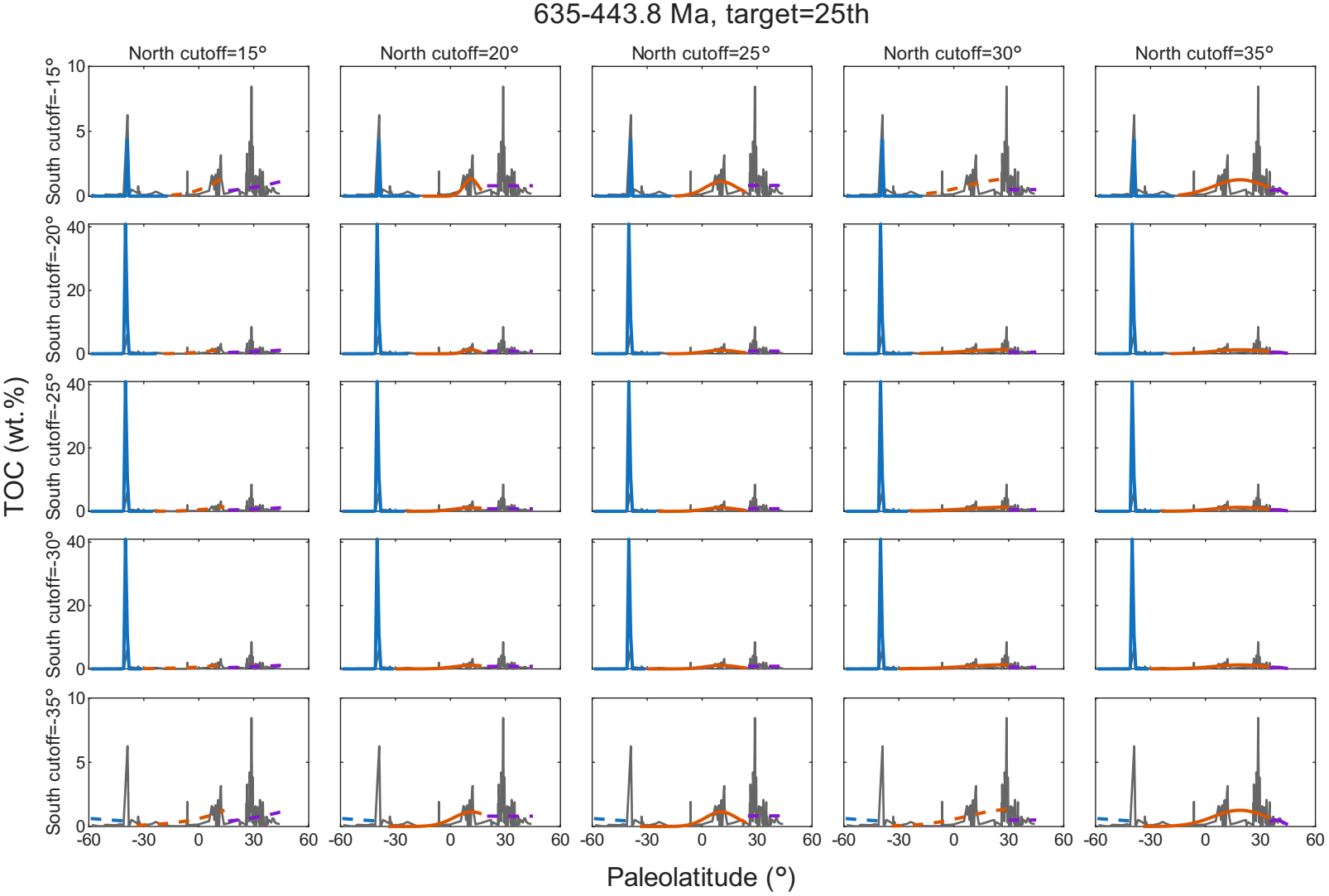

635-443.8 Ma, bin count=150
TOC (wt.%)
Paleolatitude (°)
min-max
5th-95th
10th-90th
25th-75th
median
95th
75th
median
average
443.8-298.9 Ma, bin count=150

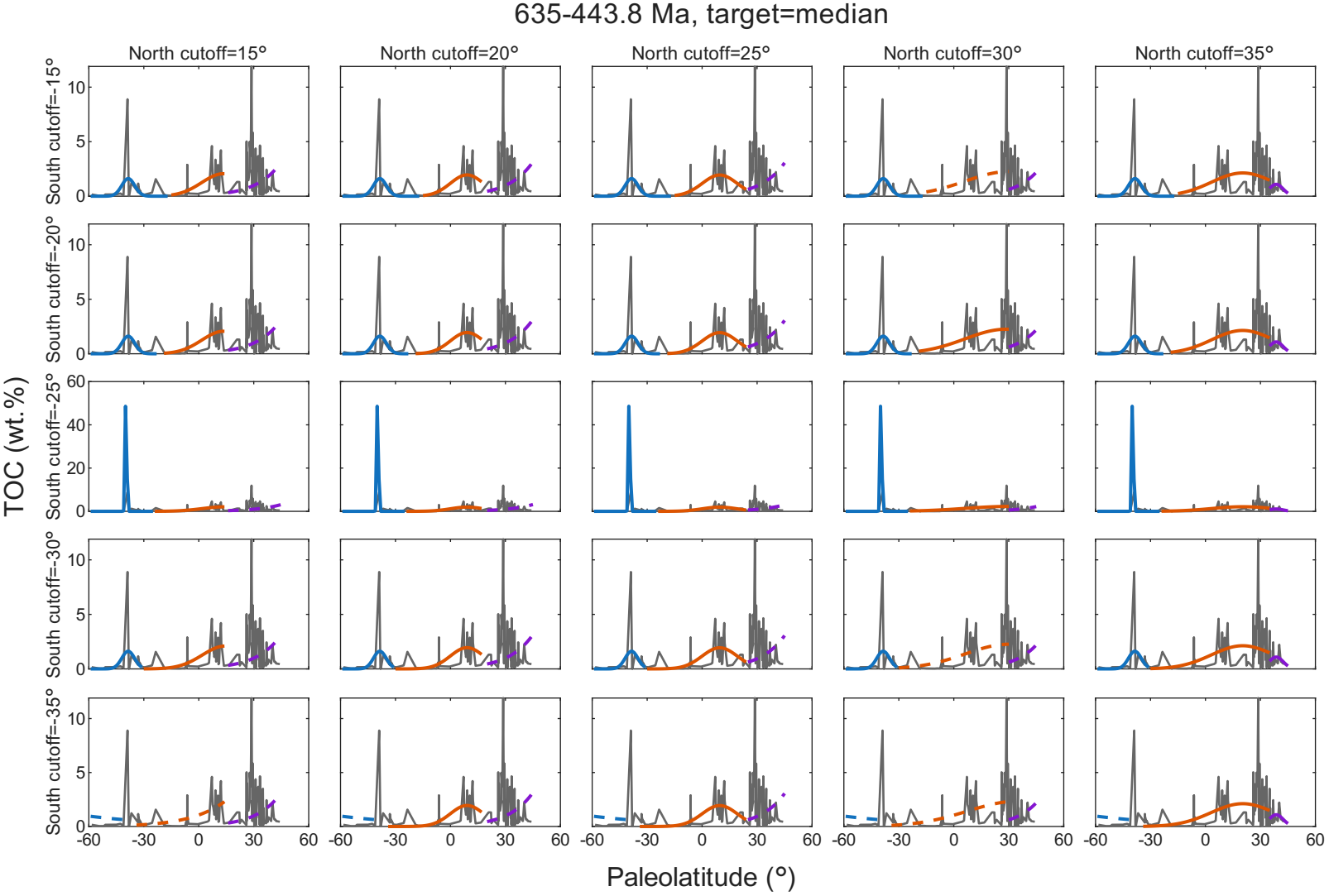

rlowess (10% window)
TOC (wt.%)
Paleolatitude (°)
10
8
6
4
2
0
-60
-40
-20
0
20
40
60
rlowess (10% window)
TOC (wt.%)
Paleolatitude (°)
25
20
15
10
5
0
-5
-60
-40
-20
0
20
40
60

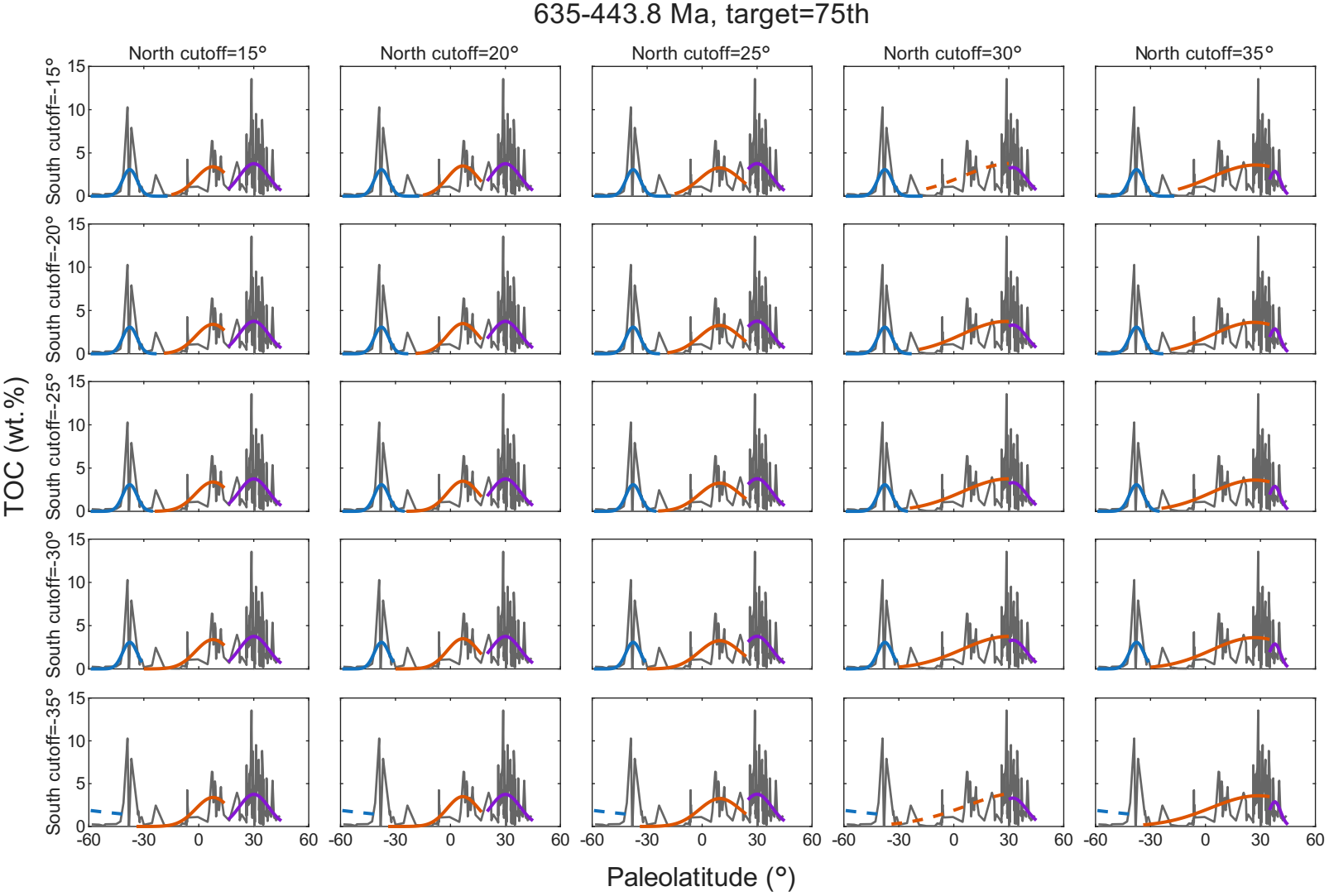

635-443.8 Ma, target=min
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
Paleolatitude (°)
0
2
4
-60
-30
0
30
60

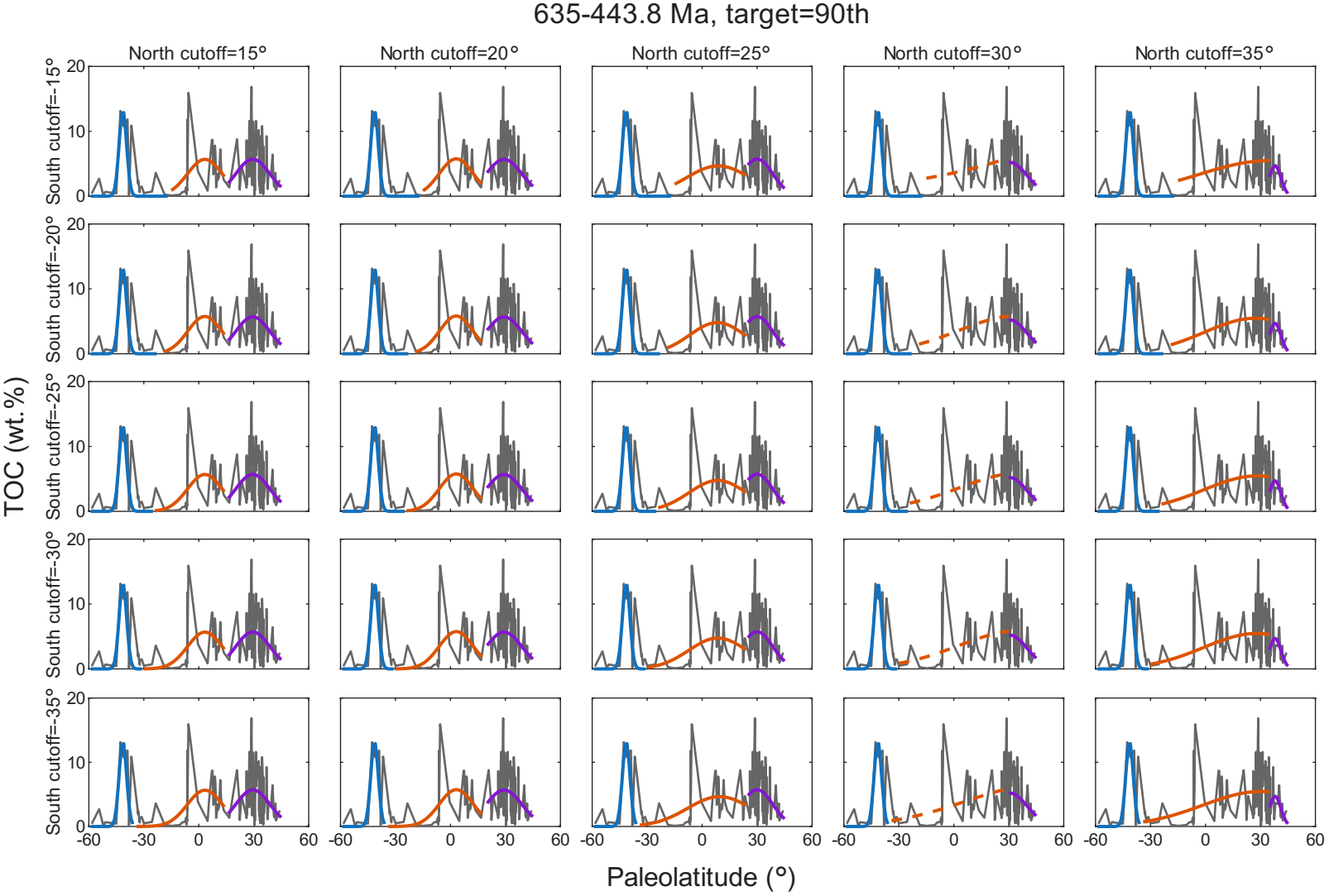

635-443.8 Ma, target=5th
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
0
2
4
6
-60
-30
0
30
60
Paleolatitude (°)

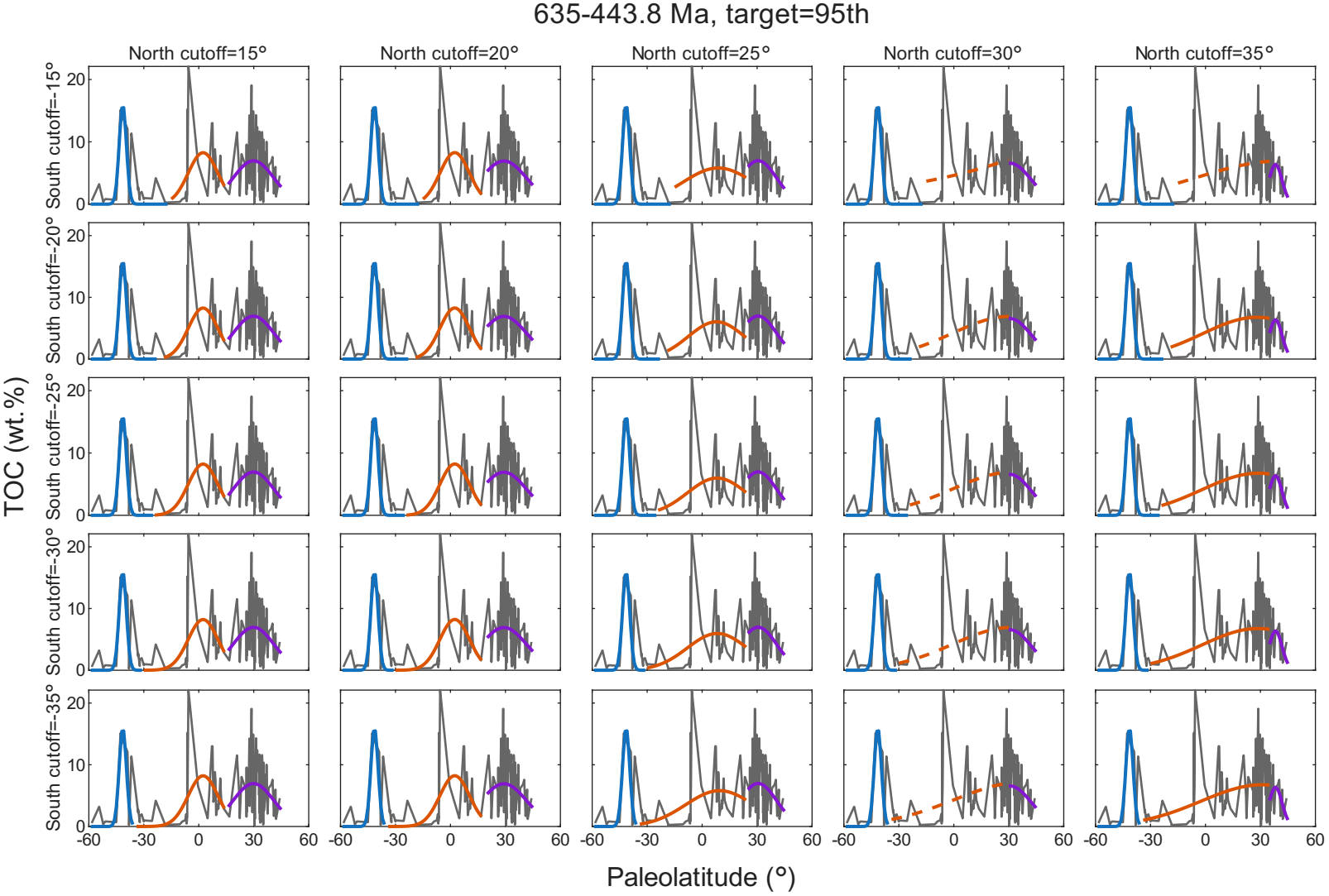

635-443.8 Ma, target=10th
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
Paleolatitude (°)
0
2
4
6
-60
-30
0
30
60

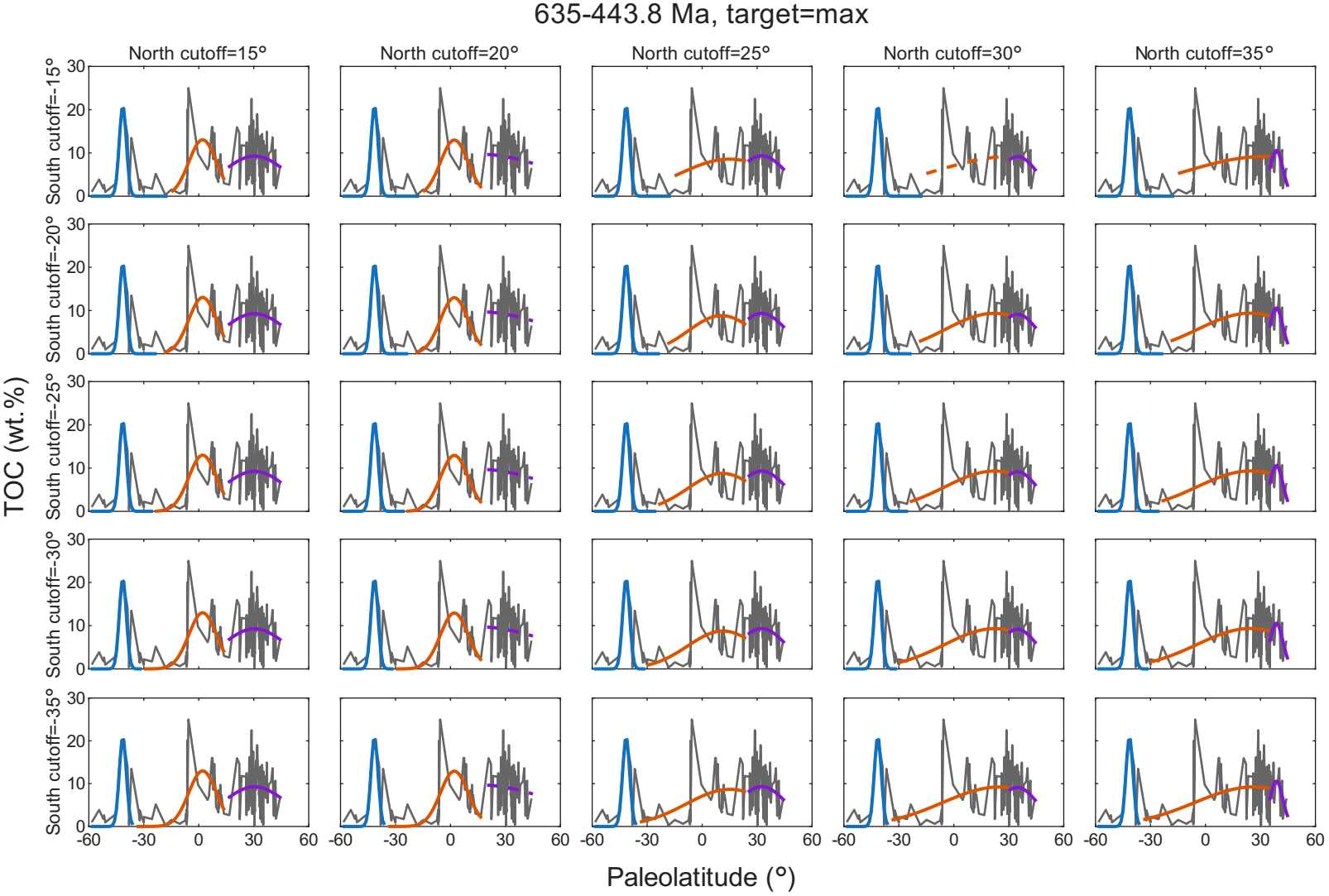

635-443.8 Ma, target=25th
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
Paleolatitude (°)
-60
-30
0
30
60
0
5
10
20
40

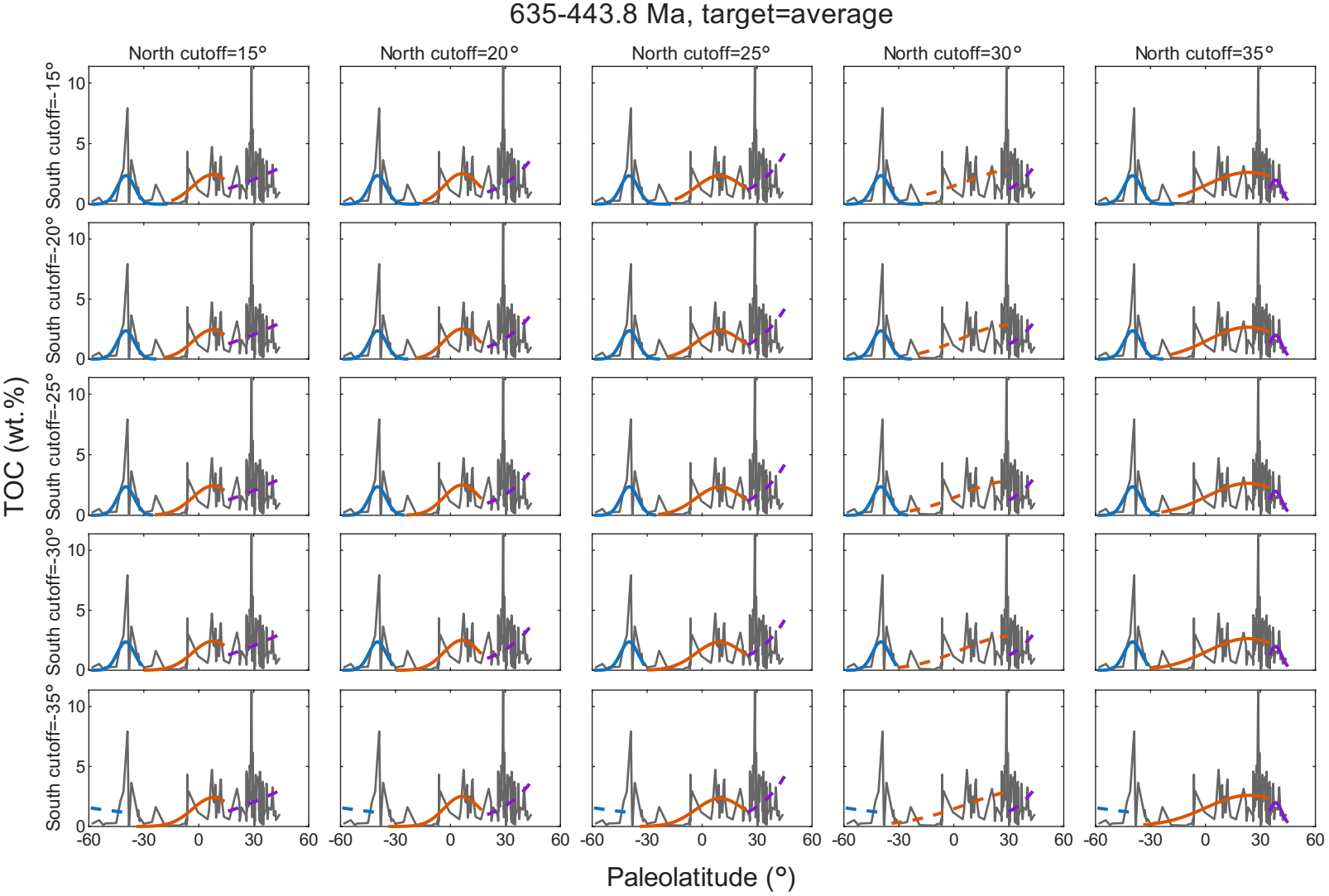

635-443.8 Ma, target=median
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
Paleolatitude (°)
-60
-30
0
30
60
0
5
10
20
40

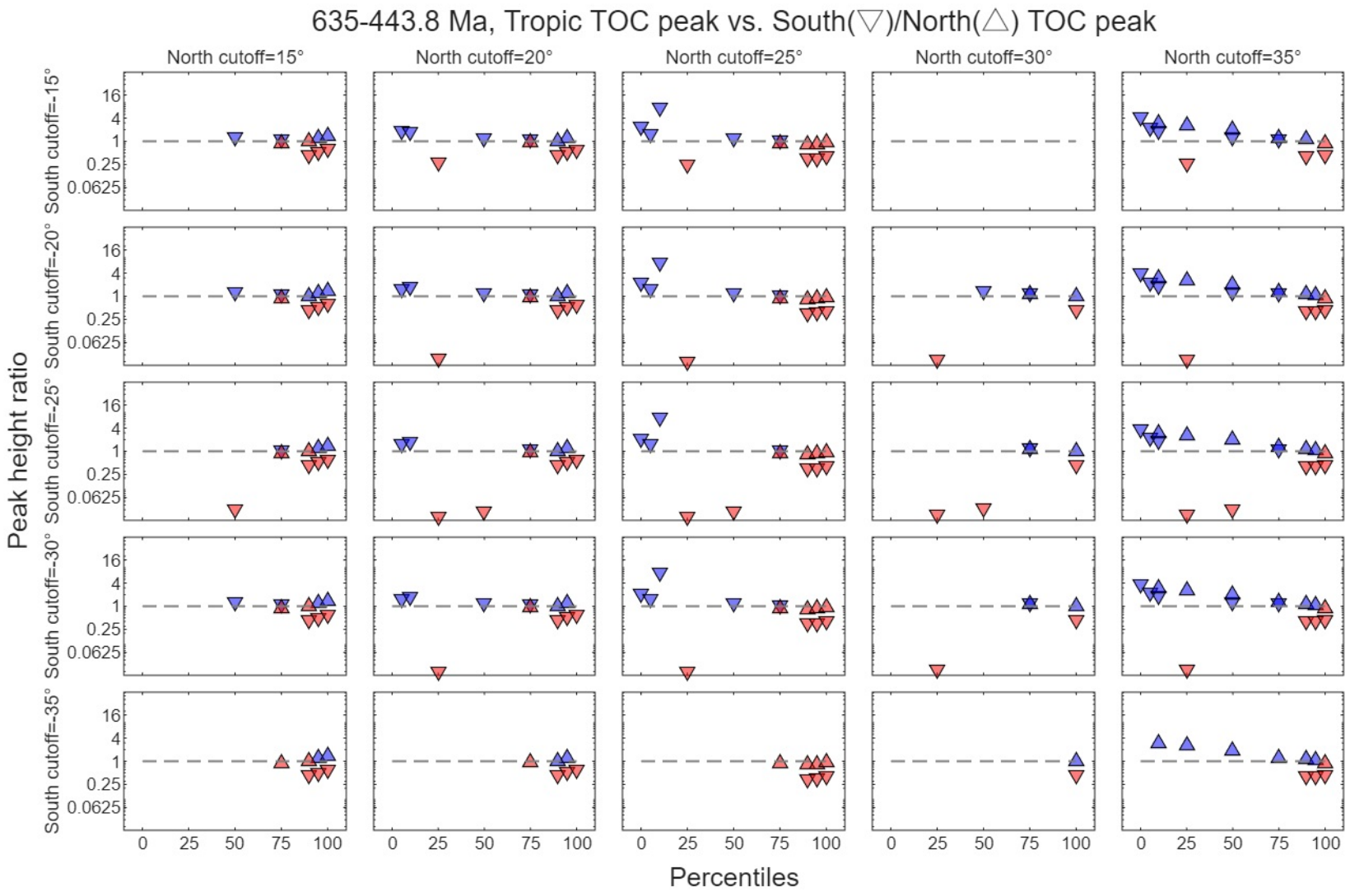

635-443.8 Ma, target=75th
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
Paleolatitude (°)
0
5
10
15
-60
-30
0
30
60

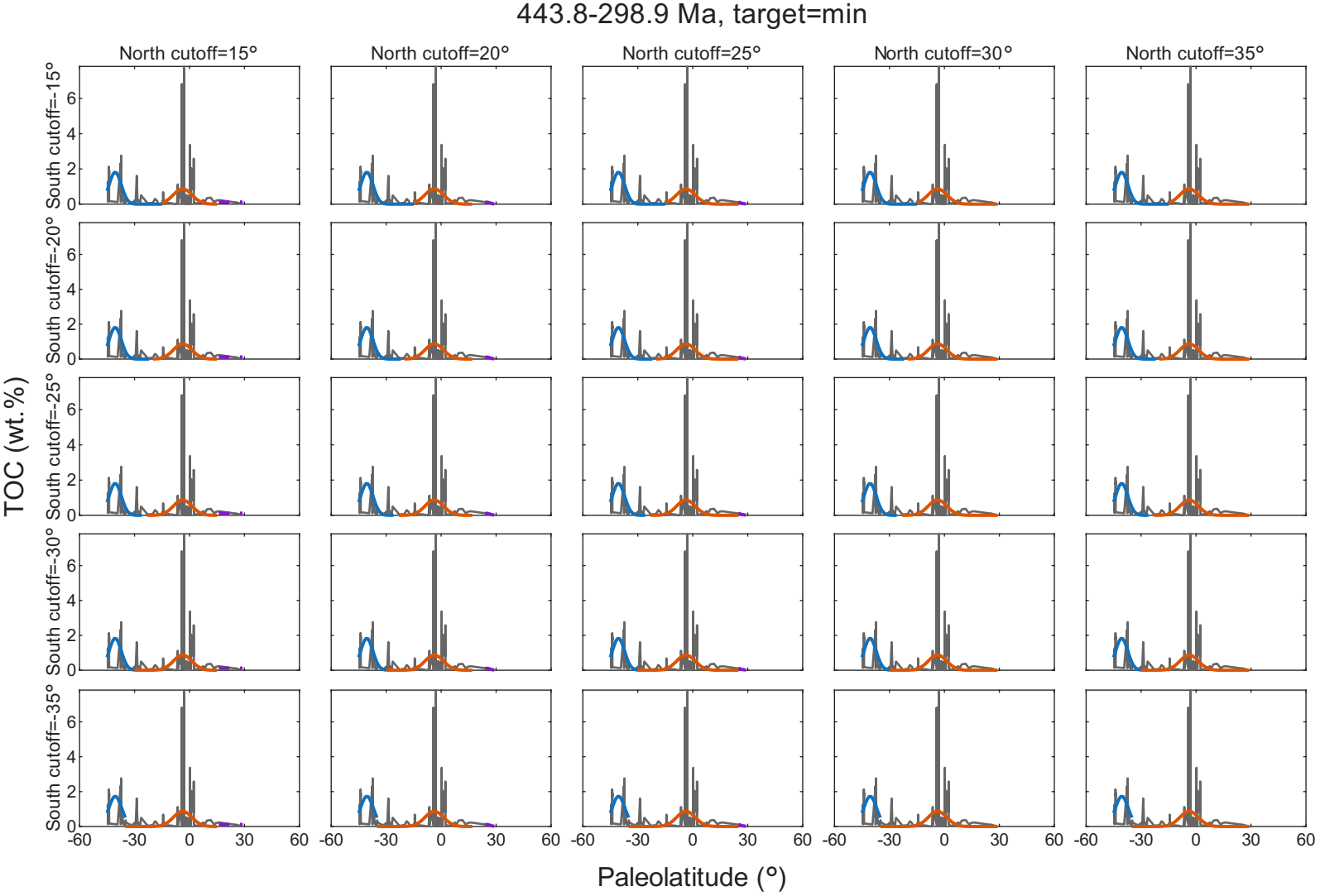

635-443.8 Ma, target=90th
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
Paleolatitude (°)
0
10
20
-60
-30
0
30
60

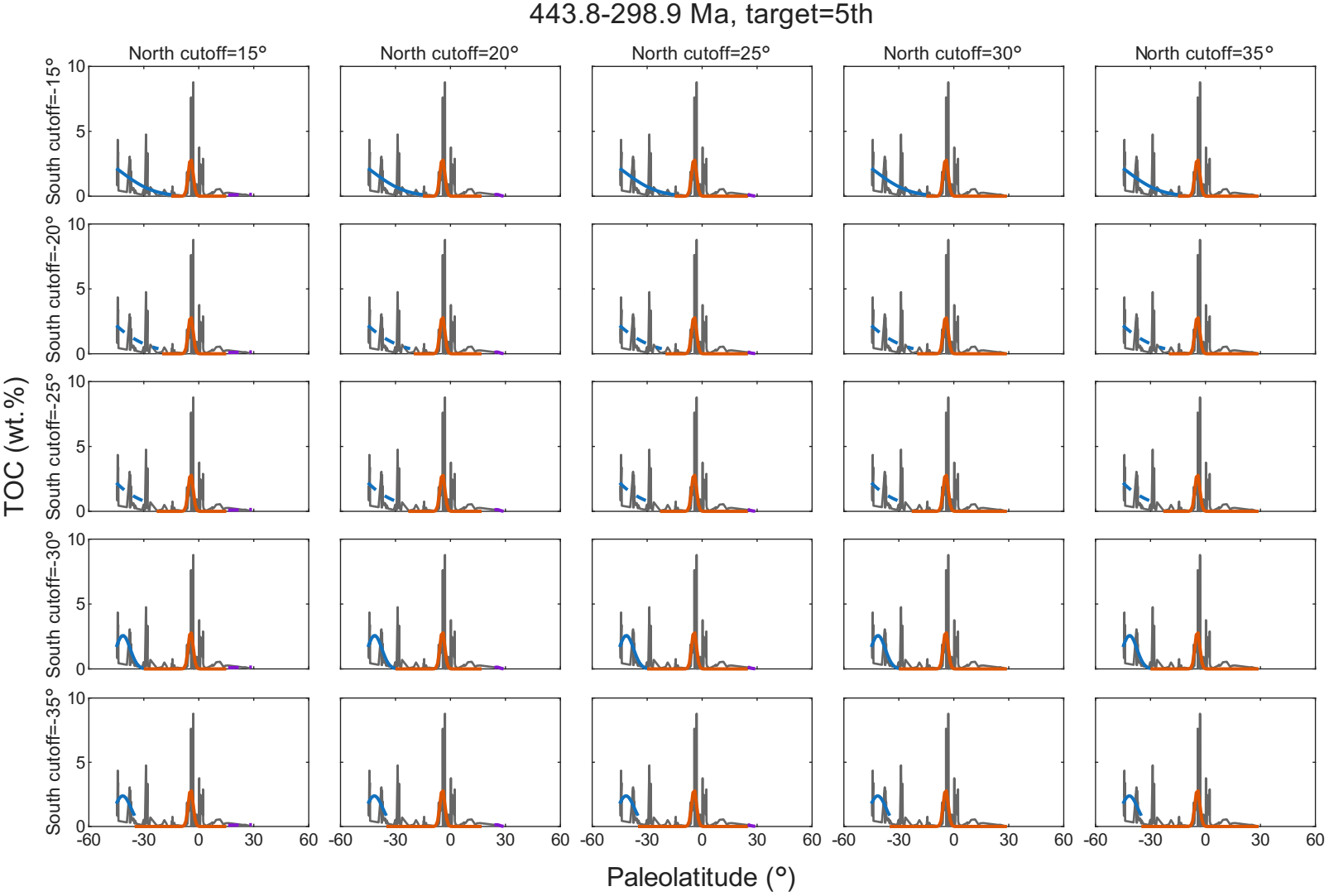

635-443.8 Ma, target=95th
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
Paleolatitude (°)
0
10
20
-60
-30
0
30
60

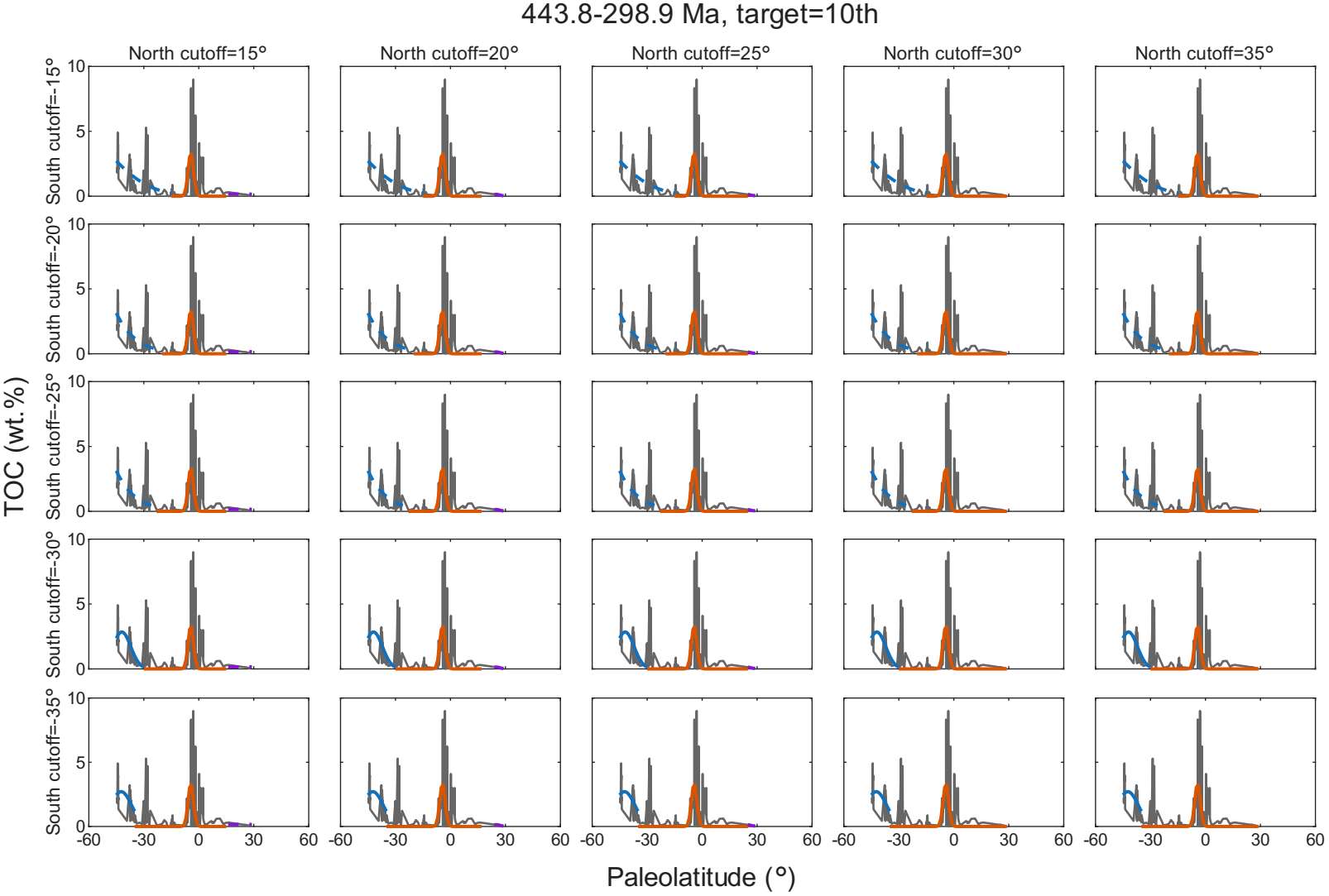

635-443.8 Ma, target=max
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
Paleolatitude (°)
0
10
20
30
-60
-30
0
30
60

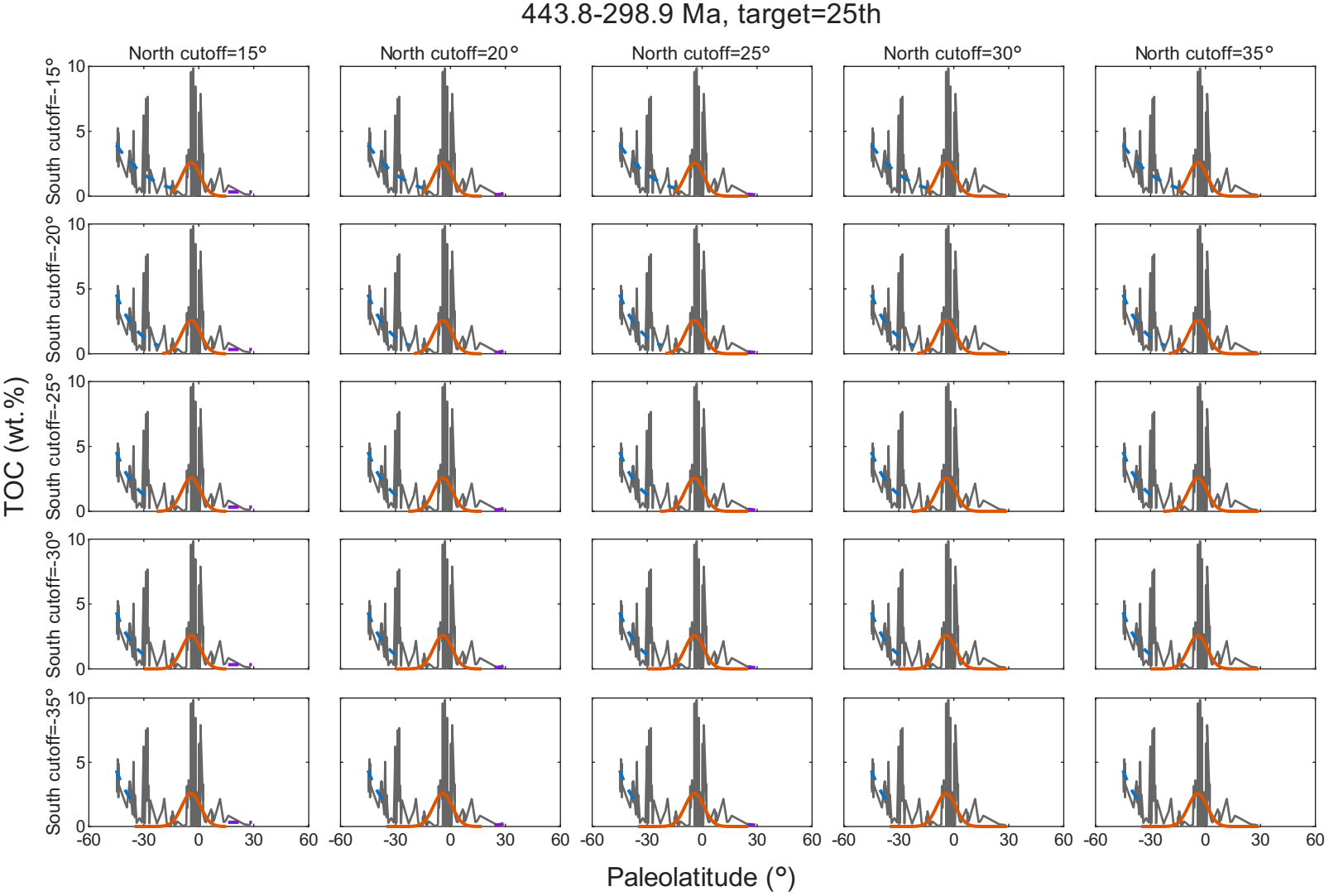

635-443.8 Ma, target=average
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
Paleolatitude (°)
0
5
10
-60
-30
0
30
60

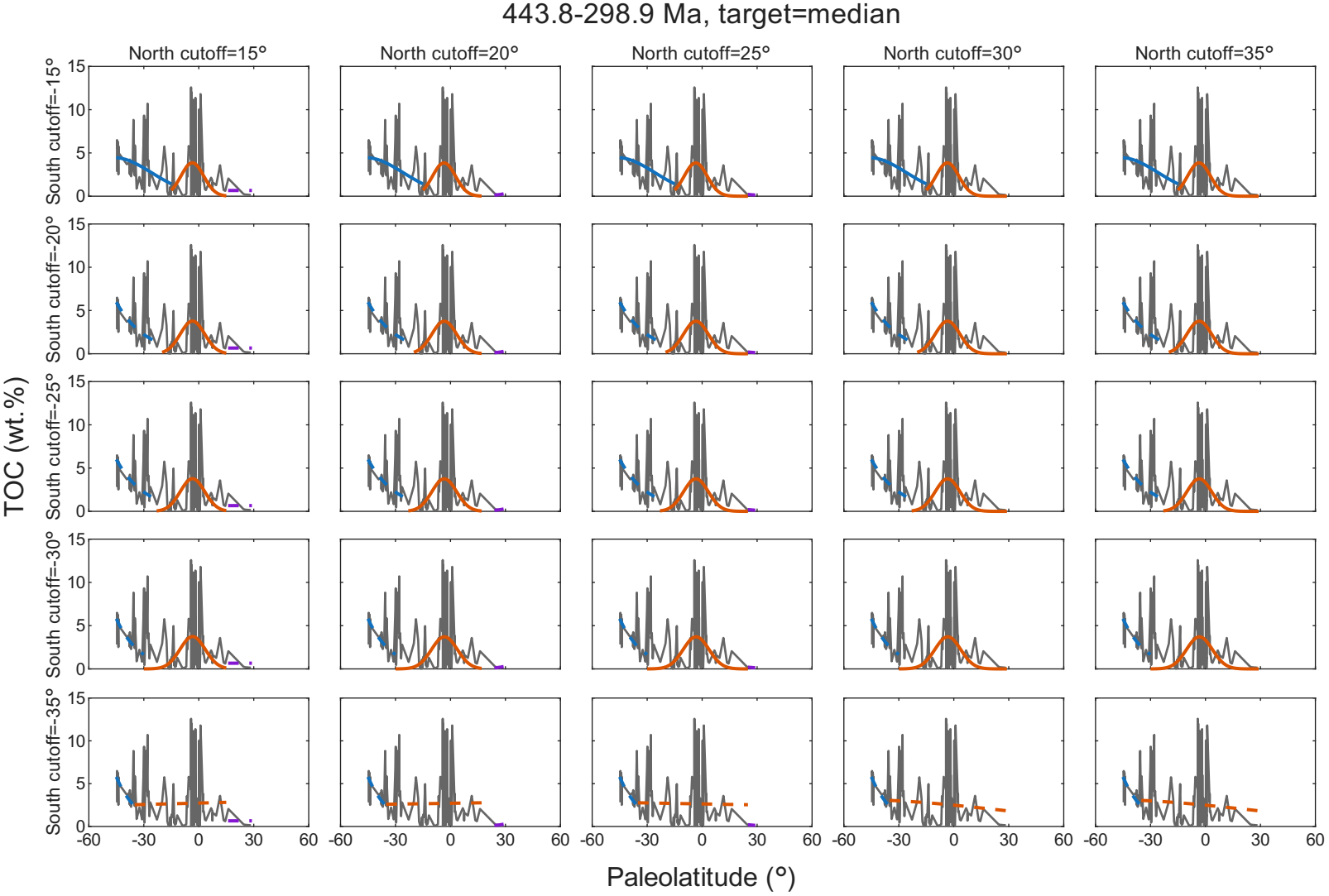

635-443.8 Ma, Tropic TOC peak vs. South(▽)/North(△) TOC peak
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
Peak height ratio
16
4
1
0.25
0.0625
0
25
50
75
100
Percentiles

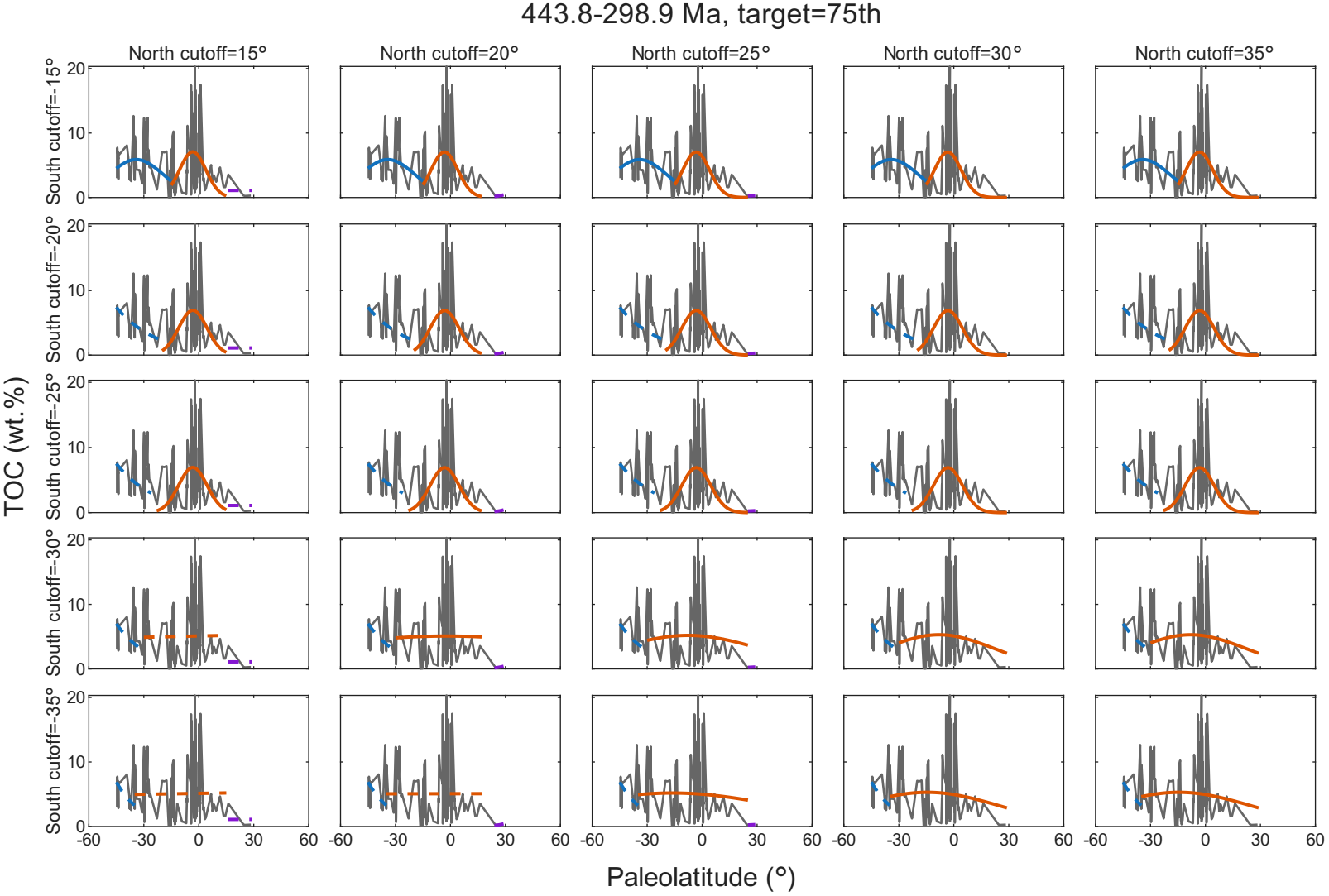

443.8-298.9 Ma, target=min
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
Paleolatitude (°)
0
2
4
6
-60
-30
0
30
60

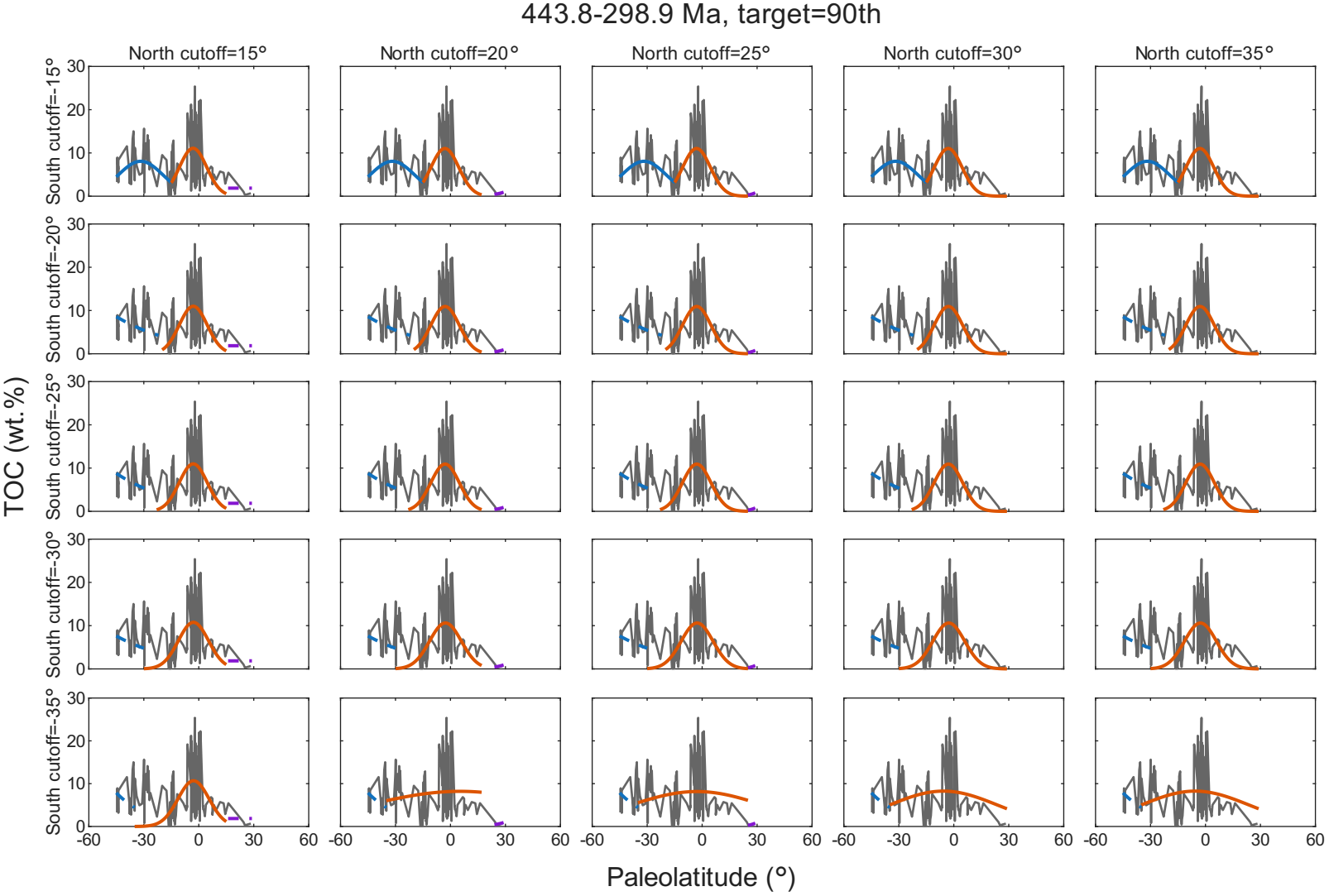

443.8-298.9 Ma, target=5th
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
Paleolatitude (°)
0
5
10
-60
-30
0
30
60

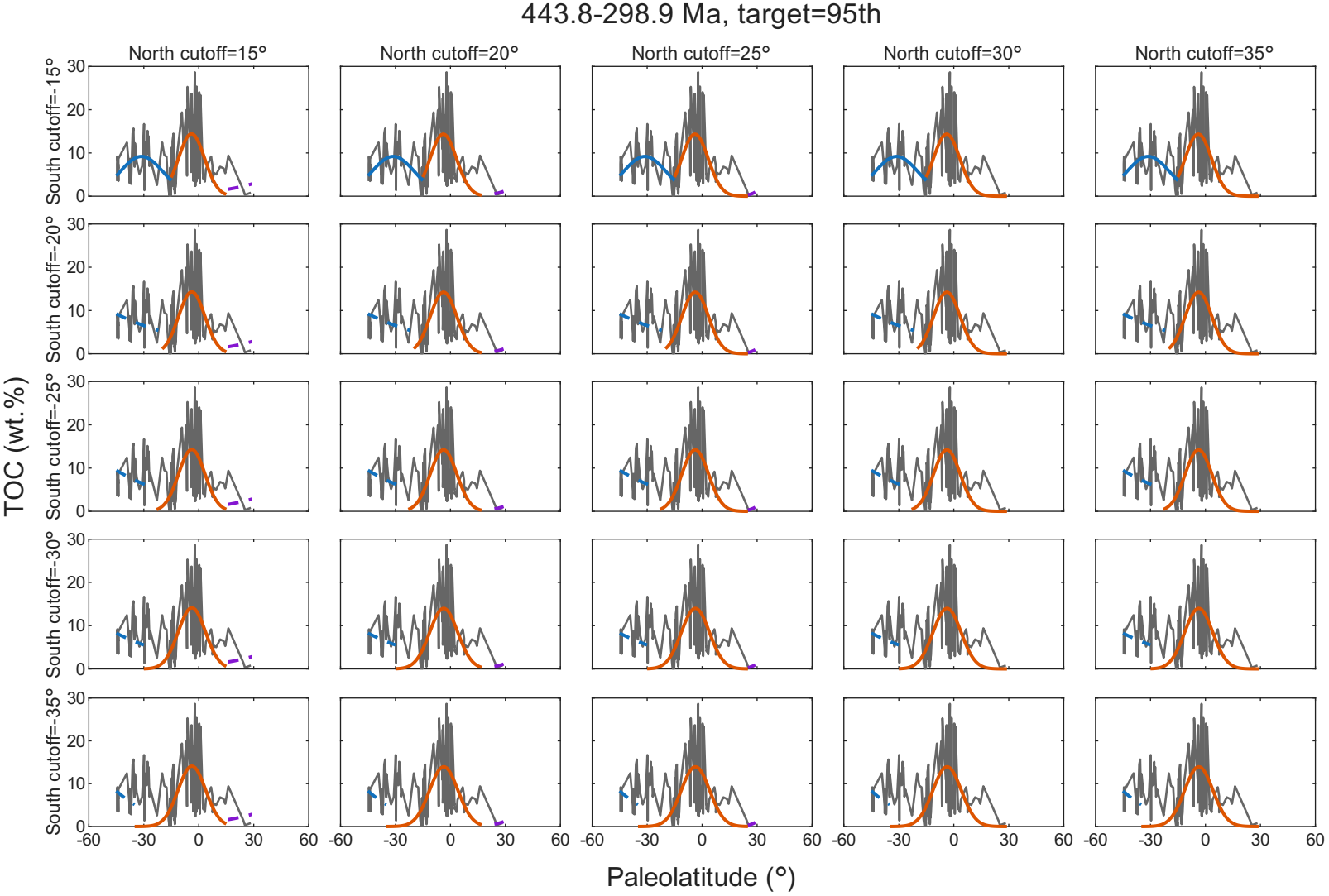

443.8-298.9 Ma, target=10th
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
Paleolatitude (°)
0
5
10
-60
-30
0
30
60

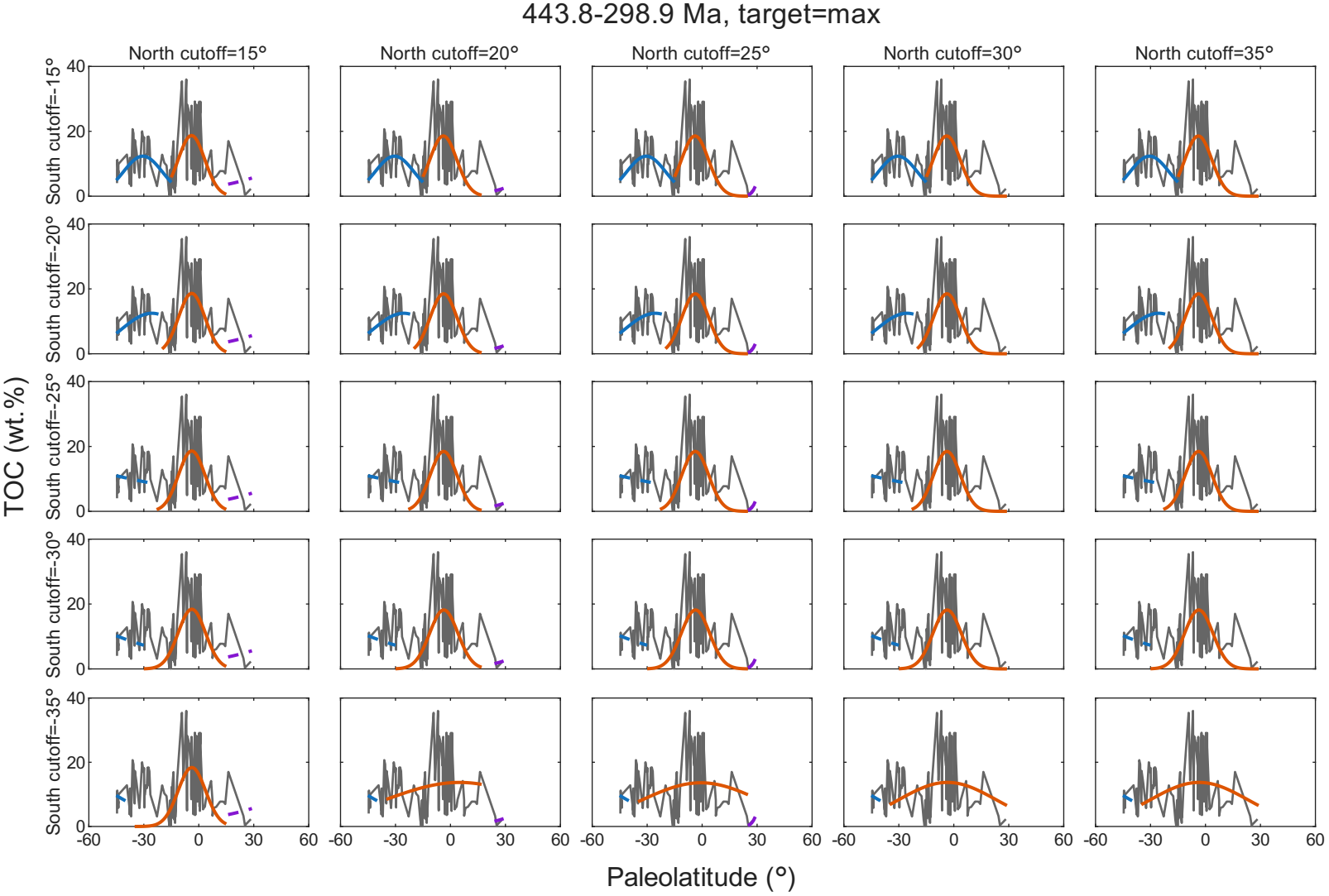

443.8-298.9 Ma, target=25th
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
Paleolatitude (°)
0
5
10
-60
-30
0
30
60

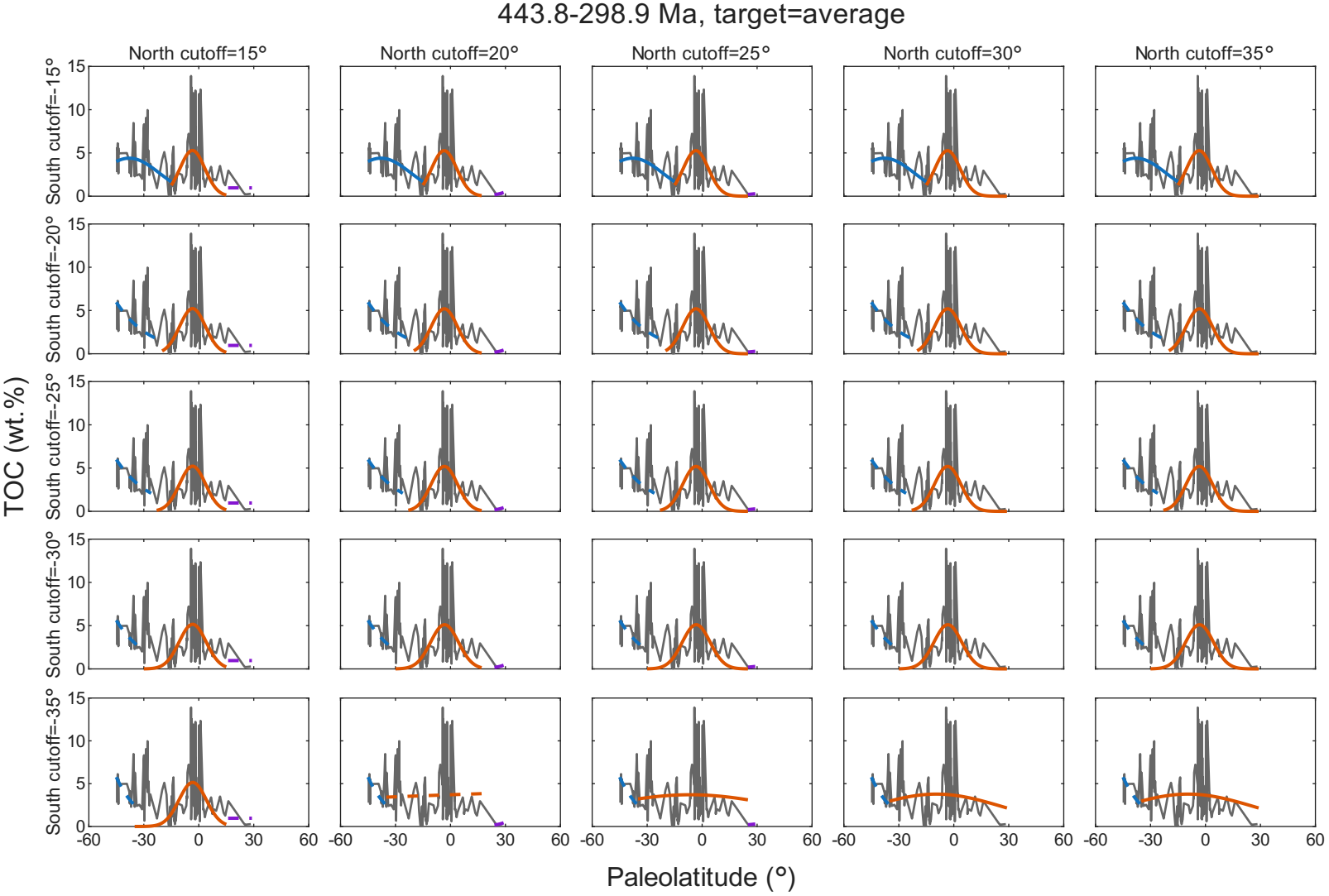

443.8-298.9 Ma, target=median
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
Paleolatitude (°)
0
5
10
15
-60
-30
0
30
60

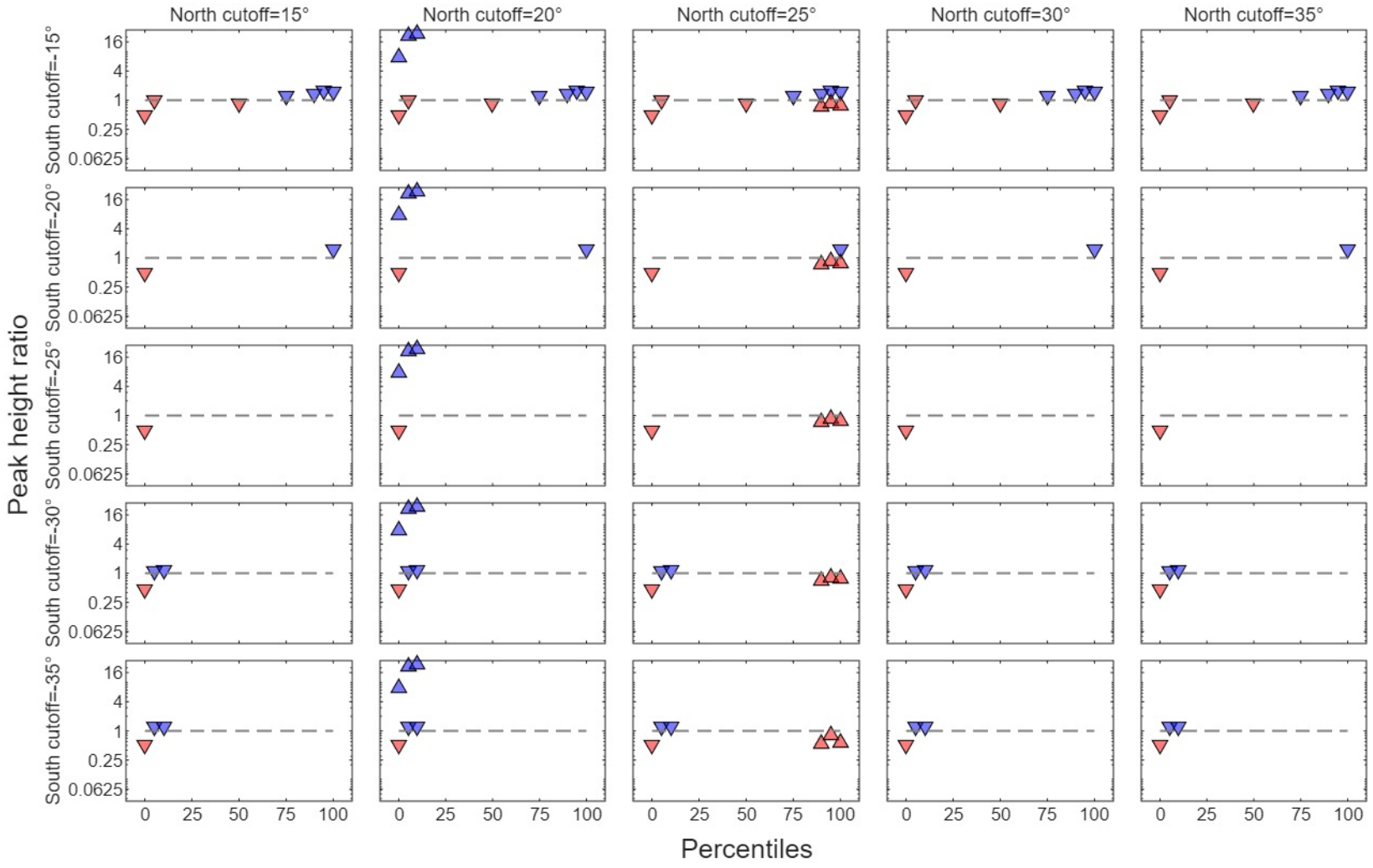

443.8-298.9 Ma, target=75th
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
Paleolatitude (°)
0
10
20
-60
-30
0
30
60


443.8-298.9 Ma, target=90th
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
Paleolatitude (°)
0
10
20
30
-60
-30
0
30
60


443.8-298.9 Ma, target=95th
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
Paleolatitude (°)
0
10
20
30
-60
-30
0
30
60


443.8-298.9 Ma, target=max
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
Paleolatitude (°)
0
20
40
-60
-30
0
30
60


443.8-298.9 Ma, target=average
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
TOC (wt.%)
Paleolatitude (°)
0
5
10
15
-60
-30
0
30
60


443.8-298.9 Ma, Tropic TOC peak vs. South(▽)/North(△) TOC peak
North cutoff=15°
North cutoff=20°
North cutoff=25°
North cutoff=30°
North cutoff=35°
South cutoff=-15°
South cutoff=-20°
South cutoff=-25°
South cutoff=-30°
South cutoff=-35°
Peak height ratio
16
4
1
0.25
0.0625
0
25
50
75
100
Percentiles